B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

IT 19-204-0406 OBJECT ORIENTED PROGRAMMING IN C++

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the basic concepts of object-oriented programming.

CO2: Describe the procedural and object oriented paradigm with concepts of streams, classes, functions, data and objects.

CO3: Implement object oriented programming constructs like encapsulation, inheritance and polymorphism.

CO4: Understand dynamic memory management techniques using pointers, constructors, destructors, etc.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	What are the benefits of using object-oriented programming over procedural programming?	3	L2	1	1,2
	(b)	Describe the use of manipulators in managing output in C++.	3	L2	1	1,2,4
	(c)	Differentiate between default and parameterized constructors in C++.	3	L1	2	1,4
	(d)	Explain the concept of operator overloading with an example.	3	L3	2	1,2
	(e)	Describe the role of "this" pointer in C++.	3	L1	3	1,2,3,4
	(f)	Explain the concept of hierarchical inheritance.	3	L2	3	1,2,3
	(g)	What are the basic functions used for random access in C++ files and how are they applied for both input and output operations?	3	L4	4	1,3,2
	(h)	Explain the importance of templates in C++.	3	L4	4	1,3,2

PART B

		(4 × 12 = 48)	Marks	BL	CO	PO
II.	(a)	Discuss the various types of functions in C++ including inline functions, friend functions and function overloading. Provide examples for each.	9	L1	1	1,2,3,4
	(b)	State the use of scope resolution operator and its use in C++.	3	L2	1	1,2
		OR				
III.		Discuss the concept of function overloading in C++. How does it enhance the flexibility of function usage? Provide code examples to support your explanation.	12	L4	2	1,2,3
IV.		Describe the various types of constructors available in C++. How do they differ from each other? Provide examples for each type.	12	L4	3	1,3
		OR				
V.	(a)	Discuss the concept of operator overloading in C++. Provide examples of overloading binary operators and explain the use of friend functions in operator overloading.	8	L2	3	1,3,2
	(b)	Describe the destructor in C++. How does it differ from a constructor?	4	L1	3	1

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		Question	Marks	BL	CO	PO
VI.		What are virtual base classes in C++? Discuss their purpose and implementation with suitable examples.	12	L3	4	1,3,12
		OR				
VII.	(a)	Explain the concept of inheritance in C++. Describe single inheritance, multilevel inheritance and multiple inheritance with examples.	8	L2	4	1,3,12
	(b)	What is multiple inheritance? Discuss its advantages and potential issues.	4	L1	4	1,3,12
VIII.		Explain the process of exception handling in C++. Discuss its importance and provide examples to illustrate its implementation.	12	L4	4	1,3,12
		OR				
IX.	(a)	What are file modes in C++? Discuss the different file modes available and their usage with examples.	8	L1	4	1,3,12
	(b)	Describe the steps involved in performing file operations in C++. Explain how to open, read, write and close a file.	4	L2	4	1,3,12

Bloom's Taxonomy Levels
L1 – 30%, L2 – 35%, L3 – 10%, L4 – 25%.

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